

Dong-Kyum Kim

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Research Interests

My research combines AI, physics, and neuroscience to uncover the mechanisms of learning and memory in large language models. Using representation geometry and causal interventions across mechanistic interpretability, model editing, and machine unlearning, I study how models store, retrieve, revise, and forget knowledge, as well as why attempts to alter it fail. My training in physics guides me toward simple principles behind complex behavior, while neuroscience provides testable hypotheses about memory formation and consolidation. Building on this foundation, I aim to determine how reinforcement learning preserves existing capabilities and selectively consolidates reward-relevant experience, and to translate these mechanisms into safer, more reliable, and more efficient continual learning methods.

Education

Korea Advanced Institute of Science and Technology (KAIST) 2016–2022

Ph.D. in Physics

Advisor: Prof. [Hawoong Jeong](#)

Dissertation: *Nonequilibrium Statistical Physics Study using Deep Learning*

Seoul National University (SNU)

2011–2015

B.S. in Physics, Minor in Computer Science & Engineering

Research Experience

Postdoctoral Researcher, Max Planck Institute for Security and Privacy (MPI-SP) Oct. 2024–Present

- Hosted by Prof. [Meeyoung Cha](#) (Scientific Director)
- Data Science for Humanity Group

Postdoctoral Researcher, Institute for Basic Science (IBS)

Mar. 2022–Sep. 2024

- Hosted by Prof. [Meeyoung Cha](#) (Chief Investigator)
- Data Science Group, Center for Mathematical and Computational Science

Data Science Intern, Samsung Electronics

Sep.–Dec. 2017

- Built distributed ML systems with [Daniel Kim](#) (Senior Data Scientist): multi-GPU anomaly image classification (Keras & Spark) and an Elasticsearch-based image search framework

Selected Publications

[†] denotes equal contribution.

- [S1] **Dong-Kyum Kim**[†], Youngkyoung Bae[†], Sangyun Lee, and Hawoong Jeong. “[Learning Entropy Production via Neural Networks](#).” *Phys. Rev. Lett.* **125**, 140604 (2020).
- [S2] **Dong-Kyum Kim**[†], Jea Kwon[†], Meeyoung Cha, and C. Justin Lee. “[Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity](#).” *NeurIPS* 2023.
- [S3] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “[Spontaneous emergence of rudimentary music detectors in deep neural networks](#).” *Nature Communications* 2024.
- [S4] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “[Bilinear representation mitigates reversal curse and enables consistent model editing](#).” *ICLR* 2026.
- [S5] Jea Kwon[†], **Dong-Kyum Kim**[†], Jiwon Kim[†], Yonghyun Kim, Woong Kook, and Meeyoung Cha. “[AI Engram: In Search of Memory Traces in Artificial Intelligence](#).” *ICML* 2026 (Oral).

Teaching Experience

Computational Physics Course, KAIST (Daejeon, Korea)

May 1, 2023

“*Deep learning applications: Nonequilibrium statistical physics study using AI*”

Computational Physics Course, KAIST (Daejeon, Korea)

Apr. 24, 2023

“*Deep Learning Introduction*”

Talks

Spark(l)ing Science, MPI-SP (MB/1 Seminar Room + Zoom) <i>“AI research using agents and notebooks”</i>	Jul. 24, 2026
CREx #23 (Online) <i>“Bilinear relational structure fixes reversal curse and enables consistent model editing”</i>	Feb. 24, 2026
IBS Winter School on AI-Boosted Basic Science (Daejeon, Korea) <i>“Resemblances between Transformer’s Nonlinearity and NMDA Receptor Dynamics”</i>	Dec. 13, 2022
KIAS CAINS Summer Workshop (Jeju, Korea) <i>“Working and reference memory in transformers on a navigation task”</i>	Sep. 2, 2022
KIAS Nonequilibrium Statistical Physics of Complex Systems (Seoul, Korea) <i>“Deep reinforcement learning for optimal mechanism in active Brownian particles”</i>	Jul. 25, 2022
KIAS AI and Natural Sciences Seminar (Seoul, Korea) <i>“Nonequilibrium Statistical Physics Study Using Deep Learning”</i>	Apr. 27, 2022
APCTP Workshop for Physics and Machine Learning (Jeju, Korea) <i>“Exploring optimal mechanisms in active Brownian particles via deep reinforcement learning”</i>	Nov. 26, 2021
Seoul National University Statistical Physics Seminar (Online) <i>“Methods of estimating entropy production”</i>	Feb. 1, 2021
Korean Physical Society Fall Meeting (Online) <i>“Deep reinforcement learning for feedback-controlled flashing ratchets”</i>	Nov. 6, 2020
NetSci2020 (Online) <i>“Discovering wiring patterns of neural networks via backboning”</i>	Sep. 22, 2020
Korean Physical Society Spring Meeting (Online) <i>“Neural estimator for entropy production”</i>	Jul. 13, 2020
Quantifying Success Satellite at NetSci2019 (Burlington, Vermont, USA) <i>“Quantifying Individual Reputation in Large-scale Historical Documents”</i>	May 27, 2019

Publications

[†] denotes equal contribution.

- [1] Jea Kwon[†], **Dong-Kyum Kim**[†], Jiwon Kim[†], Yonghyun Kim, Woong Kook, and Meeyoung Cha. “[AI Engram: In Search of Memory Traces in Artificial Intelligence](#).” *ICML 2026 (Oral)*.
- [2] Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. “[How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models](#).” *ACL 2026*.
- [3] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “[Bilinear representation mitigates reversal curse and enables consistent model editing](#).” *ICLR 2026*.
- [4] Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. “[Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning](#).” *ICLR 2026*.
- [5] Jea Kwon, Jiwon Kim, **Dong-Kyum Kim**, and Meeyoung Cha. “[Moir: Let the Model Direct Its Own Story for Robust Cross-Domain Knowledge Editing](#).” *ICML 2026 Workshop on Mechanistic Interpretability*.
- [6] Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. “[In Search of the Engram in LLMs](#).” *ICLR 2025 Blogpost*.
- [7] Gwangsung Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “[Spontaneous emergence of rudimentary music detectors in deep neural networks](#).” *Nature Communications* 2024.
- [8] Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. “[SUBTLE: An unsupervised platform with temporal link embedding that maps animal behavior](#).” *IJCV* (2024).
- [9] **Dong-Kyum Kim**[†], Jea Kwon[†], Meeyoung Cha, and C. Justin Lee. “[Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity](#).” *NeurIPS* 2023.

- [10] Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. “[Multidimensional entropic bound](#).” *Phys. Rev. Research* **5**, 013194 (2023).
- [11] **Dong-Kyum Kim**[†], Sangyun Lee[†], and Hawoong Jeong. “[Estimating entropy production with odd-parity state variables via ML](#).” *Phys. Rev. Research* **4**, 023051 (2022).
- [12] Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “[Inferring dissipation maps from videos using CNNs](#).” *Phys. Rev. Research* **4**, 033094 (2022).
- [13] **Dong-Kyum Kim** and Hawoong Jeong. “[Deep reinforcement learning for feedback control in a collective flashing ratchet](#).” *Phys. Rev. Research* **3**, L022002 (2021).
- [14] **Dong-Kyum Kim**[†], Youngkyoung Bae[†], Sangyun Lee, and Hawoong Jeong. “[Learning Entropy Production via Neural Networks](#).” *Phys. Rev. Lett.* **125**, 140604 (2020).
- [15] **Dong-Kyum Kim**[†], Byunghwee Lee[†], Daniel Kim, and Hawoong Jeong. “[Multi-label classification of historical documents using hierarchical attention](#).” *J. Korean Phys. Soc.* **76**, 368 (2020).

Technical Skills

- **Daily driver:** Currently running multiple Claude Code sessions (Max plan, \$200/mo) via tmux as my primary research workflow.
- **ML/Interp:** PyTorch, JAX, TransformerLens, & nnsight.
- **Infrastructure:** Distributed training & inference (Slurm system with hundreds of H100s), large-scale data pipelines

Academic Service

Program Committee: AAAI 2027, CIKM 2026.

Reviewer: TMLR, NeurIPS 2026, ICML 2026 Workshop on Mechanistic Interpretability, ICML 2026 (Received Gold Reviewer Award), ICLR 2026, NeurIPS 2025, ICML 2025 Workshop World Models, ICLR 2025 Workshop Re-Align, NeurIPS 2024 Workshop Behavioral ML, ICML 2024 Workshop LLMs and Cognition.

Awards

Pre-doctoral Fellow of Physics, KAIST 2021

In the Press

- **Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity** (*NeurIPS* 2023): [IBS Research News \(English\)](#), [IBS Research News \(Korean\)](#), [Donga Science](#), and [EurekAlert!](#).
- **Learning Entropy Production via Neural Networks** (*Phys. Rev. Lett.* **125**, 140604, 2020): [Physics and High Technology](#).

References

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